

HypatiaX — repository audit & consolidation guide

Scope: Cross-analysis of `LLM-HypatiaX-PAPERS-Public` (455 files, 106 dirs), `LLM-HypatiaX-PAPERS` (reprod), and `LLM-HypatiaX-Colab`. Goal: one clean public repo with local / GitHub CI / AWS all supported from the same codebase.

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1. Summary scorecard

Category	Count	Action
Security risks	4	Purge from git history immediately
Duplicate entry points	8	Keep 1, delete 7
Duplicate experiment dirs	2 extra copies	Delete, keep <code>hypatiax/experiments/benchmarks/</code>
CI workflows in wrong location	4 files	Move to <code>.github/workflows/</code>
Protocol shadow copies	2+ copies each	Consolidate to <code>hypatiax/protocols/</code>
Committed generated outputs	~50 files	Remove + gitignore
Deprecated / unused code	1 full dir + 10 files	Delete
Stale editor artefacts	2	Delete + gitignore
Files missing from reprod	6	Add

2. Security — act first

Do these before any other step. Deleting files is not enough — they must be removed from git history.

Files to purge

File	Location	Risk
<code>secrets.py</code>	reprod root	Likely contains API keys
<code>secrets-old.py</code>	reprod root	Previous key version
<code>secrets-copy.txt</code>	reprod root	Plaintext copy of keys
<code>config_secrets.py</code>	<code>hypatiax/</code>	May import or reference keys

How to purge

```
bash
```

```
# Install git-filter-repo (preferred over git filter-branch)
pip install git-filter-repo

# Run from repo root — repeat for each file
git filter-repo --path secrets.py --invert-paths
git filter-repo --path secrets-old.py --invert-paths
git filter-repo --path secrets-copy.txt --invert-paths

# For config_secrets.py: audit first
grep -n "key\|token\|secret\|password\|api" hypatiax/config_secrets.py
# If it contains hardcoded values → purge; if it only reads from env → keep but audit
```

After purging

```
bash
```

```
# Force-push all branches (coordinate with collaborators)
git push origin --force --all
git push origin --force --tags

# Rotate any exposed keys immediately via their provider dashboard
```

Prevention

Add to `.gitignore`:

```
secrets*.py
secrets*.txt
*.env
config_secrets.py    # if purged
```

3. Duplicate entry points

Problem: The repo has 8+ versions of `run_all` — the canonical entrypoint is ambiguous.

File	Location	Action
<code>run_all.sh</code>	<code>./</code>	Keep — canonical
<code>run_all.sh</code>	<code>hypatiax-all-cloud/</code>	Delete
<code>run_all.sh</code>	<code>hypatiax/RUN_ALL_OLD/</code>	Delete (entire <code>RUN_ALL_OLD/</code> dir)
<code>run_all.sh</code>	<code>hypatiax-all-checkpoint/ RUN_ALL/</code>	Delete
<code>run_all_checkpoint.py</code>	<code>./</code>	Delete
<code>run_all_injected.sh</code>	<code>hypatiax/RUN_ALL_OLD/</code>	Delete
<code>run_all_orchestrator.sh</code>	<code>hypatiax-all-checkpoint/ RUN_ALL_INJECTED/</code>	Delete
<code>repro_master.py</code>	<code>./</code>	Delete — superseded by CI
<code>recover_failed_experiments.py</code>	<code>./</code>	Delete — one-time script

Clean up

```
bash

rm -rf hypatiax/RUN_ALL_OLD/
rm -rf hypatiax-all-checkpoint/RUN_ALL/
rm -rf hypatiax-all-checkpoint/RUN_ALL_INJECTED/
rm   hypatiax-all-cloud/run_all.sh
rm   run_all_checkpoint.py
rm   repro_master.py
rm   recover_failed_experiments.py
```

4. Duplicate experiment directories

Problem: The same 10-11 experiment scripts exist in three locations.

Directory	Files	Action
<code>hypatiax/experiments/benchmarks/</code>	11 scripts	Keep — canonical
<code>hypatiax-all-checkpoint/experiments/benchmarks/</code>	11 scripts (identical)	Delete entire dir
<code>hypatiax-all-cloud/experiments/</code>	10 scripts (identical)	Delete entire dir

```
bash
```

```
# Verify they are truly identical before deleting
```

```
diff -rq hypatiax/experiments/benchmarks/ hypatiax-all-checkpoint/experiments/benchmarks/
```

```
diff -rq hypatiax/experiments/benchmarks/ hypatiax-all-cloud/experiments/
```

```
# Then delete
```

```
rm -rf hypatiax-all-checkpoint/experiments/
```

```
rm -rf hypatiax-all-cloud/experiments/
```

One script not yet in canonical location

`exp1_ablation.py` exists in `hypatiax-all-checkpoint/experiments/benchmarks/` and `hypatiax-all-cloud/experiments/` but **not** in `hypatiax/experiments/benchmarks/`. Promote it:

```
bash
```

```
cp hypatiax-all-checkpoint/experiments/benchmarks/exp1_ablation.py \
  hypatiax/experiments/benchmarks/
```

5. CI workflows — wrong location

Problem: All four CI workflows live inside `hypatiax-all-cloud/` instead of `.github/workflows/`. GitHub Actions does not pick them up from there.

File	Current location	Target location
ci_experiment.yml	hypatiax-all-cloud/	.github/workflows/
ci_consolidate_experiment.yml	hypatiax-all-cloud/	.github/workflows/
ci_schedule_all.yml	hypatiax-all-cloud/	.github/workflows/
dispatch_experiment.sh	hypatiax-all-cloud/	.github/scripts/
ci_trace_pipeline.yml	(new — not yet created)	.github/workflows/

```
bash
```

```
mkdir -p .github/workflows .github/scripts
```

```
mv hypatiax-all-cloud/ci_experiment.yml .github/workflows/
mv hypatiax-all-cloud/ci_consolidate_experiment.yml .github/workflows/
mv hypatiax-all-cloud/ci_schedule_all.yml .github/workflows/
mv hypatiax-all-cloud/dispatch_experiment.sh .github/scripts/
# Add ci_trace_pipeline.yml (already written separately)
```

AWS CI files — retire

These were AWS/GCP experiments and are no longer needed:

```
bash
```

```
# Inside hypatiax-all-aws/
rm ci.yml ci-aws.yml ci-gcp.yml ci-change-arch2.yml buildspec.yml
```

6. Protocols — shadow copies

Problem: Protocol files exist in three locations simultaneously.

Location	Files	Action
<code>hypatiaux/protocols/</code>	3 files (all_30, benchmark_v2, defi)	Keep — canonical
<code>hypatiaux-all-checkpoint/protocols/</code>	14 files (full set)	Merge missing ones up, then delete dir
<code>hypatiaux-all-cloud/protocols/</code>	3 files (same as hypatiaux/protocols/)	Delete dir
<code>no-used-hypatiaux/protocols/</code>	7 deprecated versions	Delete dir

Merge and consolidate

```

bash

# Copy the 11 files missing from hypatiaux/protocols/ (from checkpoint)
MISSING=(
  _base.py
  experiment_protocol_ablation_exp1.py
  experiment_protocol_defi_v3.py
  experiment_protocol_extrapolation_comparative.py
  experiment_protocol_feynman_exp2.py
  experiment_protocol_hybrid_routing.py
  experiment_protocol_instability_rf02_04.py
  experiment_protocol_nguyen12_exp3.py
  experiment_protocol_noise_sweep.py
  experiment_protocol_provenance_audit.py
  universal_protocol.py
)
for f in "${MISSING[@]"; do
  cp hypatiaux-all-checkpoint/protocols/$f hypatiaux/protocols/
done

# Delete shadow copies
rm -rf hypatiaux-all-checkpoint/protocols/
rm -rf hypatiaux-all-cloud/protocols/
rm -rf no-used-hypatiaux/protocols/

```

7. Output generators

File	Location	Action
<code>generate_figures.py</code>	<code>figures/</code>	Keep — canonical
<code>generate_figures.py</code>	<code>scripts/patches/</code>	Delete — duplicate
<code>generate_tables.py</code>	<code>tables/</code>	Keep — canonical
<code>generate_tables.py</code>	<code>scripts/patches/</code>	Delete — duplicate

```
bash
```

```
rm scripts/patches/generate_figures.py
rm scripts/patches/generate_tables.py
```

scripts/patches/ — what to keep

File	Keep?	Reason
<code>apply_patches.py</code>	Yes	reprod utility
<code>generate_patches.py</code>	Yes	reprod utility
<code>verify_results.py</code>	Yes	reprod utility
<code>validate_code.py</code>	Yes	reprod utility
<code>clean_results.py</code>	Yes	reprod utility
<code>check_hypatiax_protocols.py</code>	Yes	pub utility
<code>patch_log.jsonl</code>	Audit	May contain sensitive paths — check before keeping
<code>pipeline--patches.txt</code>	Audit	Same
<code>generated/*.patch.json</code>	Gitignore	Generated — do not commit

8. Audit / trace scripts

Problem: Trace/provenance scripts are split between two locations and not unified.

File	Location	Action
<code>pipeline_trace.py</code>	<code>audit/</code> (public)	Base of tracer — keep
<code>discover_provenance.py</code>	<code>audit/</code> (public)	Keep
<code>provenance_audit.py</code>	<code>audit/</code> (public)	Keep
<code>scan_internal_imports.py</code>	<code>audit/</code> (public)	Keep
<code>trace_pipeline_1_.py</code>	reprod root	Move to <code>audit/</code> , rename <code>trace_pipeline.py</code>
<code>ci_trace_pipeline.yml</code>	(new)	Add to <code>.github/workflows/</code>

```
bash
```

```
mv trace_pipeline_1_.py audit/trace_pipeline.py
# Update ci_trace_pipeline.yml: TRACE_SCRIPT="audit/trace_pipeline.py"
```

9. Committed generated outputs

These files are the **product** of running experiments — they should never be in git. They bloat the repo, cause merge conflicts, and are reproducible from code.

Root-level results (delete + gitignore)

```
bash
```

```
rm exp1_ablation_checkpoint.json
rm exp1_ablation_results.json
rm exp1_ablation_table.tex
rm exp1_instability_stats.json
rm exp1_rf01_mannwhitney.json
rm instability_extrapolation_v2.csv
rm provenance_map_exp1.json
rm clean_exp2_sym_checkpoint.py # one-time cleanup script
```

hypatiax/data/results/ (gitignore entire dir)

```
bash
```

```
# Keep the directory structure, remove committed files  
git rm -r --cached hypatiax/data/results/  
# Contents are reproduced by running the pipeline
```

logs/ (gitignore)

```
bash
```

```
git rm -r --cached logs/  
# Exception: logs/README.md – keep if it has instructions
```

scripts/paper/tables/*.tex (gitignore)

```
bash
```

```
git rm --cached scripts/paper/tables/*.tex
```

scripts/patches/generated/ (gitignore)

```
bash
```

```
git rm -r --cached scripts/patches/generated/
```

10. Deprecated / unused code

`no-used-hypatiax/` — delete entire directory

The directory name is self-documenting. Contains old experiments and deprecated protocol versions that have been superseded.

```
bash
```

```
rm -rf no-used-hypatiax/
```

`hypatiax-all-checkpoint/` — delete after migration

Once protocols and experiment scripts are merged into canonical locations (sections 4 and 6), the entire dir is redundant:

```
bash
```

```
rm -rf hypatiax-all-checkpoint/
```

hypatiax-all-aws/ — archive or move

This contains legitimate AWS infrastructure docs but should not live at repo root. Options:

- Move useful files to `aws/` (docs, `merge_exp2_shards.py`, `PROD/`)
- Delete CI files already covered by `.github/workflows/`

```
bash

mkdir -p aws/docs
mv hypatiax-all-aws/AWS-GUIDE.txt      aws/docs/
mv hypatiax-all-aws/AWS-GUIDE-REVIEWER.txt aws/docs/
mv hypatiax-all-aws/budget-to-results.md aws/docs/
mv hypatiax-all-aws/merge_exp2_shards.py aws/
mv hypatiax-all-aws/PROD/              aws/prod/
rm -rf hypatiax-all-aws/
```

HypatiaX_progressive_working_colab.ipynb (root)

Working notebook committed at root — move to `notebooks/` or delete:

```
bash

mv HypatiaX_progressive_working_colab.ipynb notebooks/
```

Duplicate paper source files at root

```
bash

# paper/ dir is canonical
rm jmlr-hypatiax-paper-final.tex      # duplicate of paper/jmlr-hypatiax-paper-final.tex
rm supp_benchmark_report.tex          # duplicate of paper/supp_benchmark_report.tex
rm supp_routing_improvements.tex      # duplicate of paper/supp_routing_improvements.tex
```

11. Stale editor artefacts

Emacs autosave files were accidentally committed:

```
bash

git rm '#run_all.sh#'
git rm '#run_all.sh~'
```

Add to `.gitignore`:

```
\#*\#  
*~  
.**
```

Also remove all `__pycache__` dirs and `.pyc` files:

```
bash  
  
find . -type d -name __pycache__ -exec rm -rf {} +  
find . -name "*.pyc" -delete
```

12. Files to add to reprod

Files that exist in the public repo but are missing from the reprod repo and should be added:

File	Public location	Add to reprod at
<code>LICENSE</code>	<code>./</code>	<code>./</code>
<code>validation_paper_config.py</code>	<code>./</code>	<code>./</code>
<code>audit/pipeline_trace.py</code>	<code>audit/</code>	<code>audit/</code>
<code>audit/discover_provenance.py</code>	<code>audit/</code>	<code>audit/</code>
<code>audit/provenance_audit.py</code>	<code>audit/</code>	<code>audit/</code>
<code>audit/ scan_internal_imports.py</code>	<code>audit/</code>	<code>audit/</code>
<code>exp1_ablation.py</code>	<code>hypatiax-all- checkpoint/</code>	<code>hypatiax/experiments/ benchmarks/</code>
<code>purge_actions_cache.sh</code>	<code>./</code>	<code>.github/scripts/</code>

13. Target repo structure

LLM-HypatiaX-PAPERS-Public/

```
├─ .github/
│   └─ workflows/
│       ├── ci_experiment.yml
│       ├── ci_consolidate_experiment.yml
│       ├── ci_schedule_all.yml
│       └─ ci_trace_pipeline.yml
└─ scripts/
    ├── dispatch_experiment.sh
    ├── print_repro.py
    ├── purge_actions_cache.sh
    └─ test_key_status.py

├─ hypatiaux/                                     ← package – identical in public + reprod
│   ├── analysis/
│   ├── core/
│   │   ├── base_pure_llm/
│   │   ├── generation/
│   │   ├── runners/
│   │   └─ training/
│   ├── experiments/
│   │   └─ benchmarks/                               ← all 12 experiment scripts here only
│   ├── protocols/                                   ← all 14 protocol files here only
│   ├── reproducibility/
│   │   └─ hash_lock.py
│   ├── shared/
│   │   └─ utilities.py
│   └─ tools/

├─ audit/                                           ← trace + provenance tools
│   ├── trace_pipeline.py
│   ├── discover_provenance.py
│   ├── provenance_audit.py
│   └─ scan_internal_imports.py

├─ config/
│   ├── repro.yaml                                   ← hyperparams – single source of truth
│   ├── envs/
│   │   ├── local.yaml
│   │   ├── ci.yaml
│   │   └─ aws.yaml
│   └─ test_key_status.py

├─ figures/
│   └─ generate_figures.py
```

- └─ tables/
 - └─ generate_tables.py
- └─ scripts/
 - └─ patches/
 - └─ apply_patches.py
 - └─ generate_patches.py
 - └─ verify_results.py
 - └─ validate_code.py
 - └─ clean_results.py
 - └─ check_hypatiaux_protocols.py
- └─ notebooks/
 - └─ HypatiaX_Experiments_v7_PAPER_QUALITY.ipynb
 - └─ HypatiaX_Experiments_v7_PUBLIC_fast.ipynb
 - └─ HypatiaX_pipeline.ipynb
 - └─ NB-01..NB-06_*.ipynb
- └─ paper/
 - └─ jmlr-hypatiaux-paper-final.tex
 - └─ supp_benchmark_report.tex
 - └─ supp_routing_improvements.tex
- └─ aws/
 - └─ run_aws.sh
 - └─ merge_exp2_shards.py
 - └─ docs/
 - └─ prod/
- └─ docs/
 - └─ REPRODUCTION_GUIDE.md
 - └─ COMPATIBILITY-LIST.txt
 - └─ protocols.txt
- └─ hypatiaux/data/results/ ← gitignored, populated by CI
- └─ logs/ ← gitignored
- └─ activate_hypatiaux.sh
- └─ install.sh
- └─ run_all.sh ← single canonical entrypoint
- └─ Makefile
- └─ requirements.txt
- └─ requirements-exact.txt
- └─ pyproject.toml
- └─ setup.py
- └─ VERSION

- └─ LICENSE
- └─ README.md
- └─ REPRODUCIBILITY.md

14. .gitignore additions

Add these to `.gitignore` if not already present:

```
gitignore

# Python
__pycache__/
*.pyc
*.pyo
*.pyd
.Python

# Secrets
secrets*.py
secrets*.txt
*.env
.env.*
config_secrets.py

# Editor artefacts
\#*\#
*~
.#*
*.orig
*.bak

# Generated outputs – reproduced by pipeline
hypatiaux/data/results/
logs/*.json
logs/*.zip
logs/*.jsonl
exp1_ablation_checkpoint.json
exp1_ablation_results.json
exp1_ablation_table.tex
exp1_instability_stats.json
exp1_rf01_mannwhitney.json
instability_extrapolation_v2.csv
provenance_map_exp1.json
scripts/paper/tables/*.tex
scripts/patches/generated/
patches/generated/

# Checkpoints
*.checkpoint.json
*_checkpoint.json
*_shard*.json

# Build
*.egg-info/
dist/
```


dist/
build/

15. Action checklist

Work through these in order — security first, then structure, then cleanup.

Immediate (today)

- ☐ Audit `config_secrets.py` for hardcoded keys
- ☐ Purge `secrets.py`, `secrets-old.py`, `secrets-copy.txt` from git history
- ☐ Rotate any exposed API keys
- ☐ Add secrets patterns to `.gitignore`
- ☐ Delete Emacs autosave files (`#run_all.sh#`, `#run_all.sh#~`)

Structure (this week)

- ☐ Move CI workflows from `hypatiax-all-cloud/` to `.github/workflows/`
- ☐ Move `dispatch_experiment.sh` to `.github/scripts/`
- ☐ Add `ci_trace_pipeline.yml` to `.github/workflows/`
- ☐ Promote `exp1_ablation.py` to `hypatiax/experiments/benchmarks/`
- ☐ Consolidate all protocols into `hypatiax/protocols/`
- ☐ Move `trace_pipeline_1.py` to `audit/trace_pipeline.py`
- ☐ Create `config/envs/local.yaml`, `ci.yaml`, `aws.yaml`

Cleanup (this week)

- ☐ Delete `hypatiax-all-checkpoint/` (after migration)
- ☐ Delete `hypatiax-all-cloud/experiments/` and `protocols/` (after migration)
- ☐ Delete `no-used-hypatiax/`
- ☐ Delete duplicate `run_all*` entry points (keep only root `run_all.sh`)
- ☐ Delete duplicate paper `.tex` files at root (keep `paper/` copies)
- ☐ Remove and gitignore `scripts/patches/generate_figures.py` and `generate_tables.py`

Generated outputs (this week)

- ☐ Run `git rm -r --cached hypatiax/data/results/ logs/`
- ☐ Remove root-level result JSONs and `.tex` files
- ☐ Remove `scripts/patches/generated/` from git
- ☐ Update `.gitignore` with full list from section 14
- ☐ Commit and push `.gitignore` changes

AWS infra (next sprint)

- ☐ Consolidate `hypatiaux-all-aws/` useful files into `aws/`
- ☐ Write `aws/run_aws.sh` wrapping `run_all.sh` with S3 upload + teardown
- ☐ Delete `hypatiaux-all-aws/` after migration

Verify

- ☐ Run `trace_pipeline.py --shell run_all.sh --repo-root .` — expect 0 errors
- ☐ Run `ci_trace_pipeline.yml` via `workflow_dispatch` — expect green
- ☐ Run `python -c 'from hypatiaux.protocols.universal_protocol import *; print("OK")'`
- ☐ Run one experiment end-to-end locally: `HYPATIAX_ENV=local bash run_all.sh exp1`